

# Number of Clusters Discovery

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## Dataset and Preprocessing

The text8 dataset, consisting of 17 million tokens, was tokenized using NLTK's Punkt tokenizer and split into train (64%), validation (16%), and test (20%) sets. Preprocessing included lowercasing, removing non-ASCII, non-alphabetic, and stopwords (including numeric terms), and filtering tokens by length (3–18 characters). Lemmatization and synonym mapping were applied for consistency, and tokens with occurrences between 50 and 5,000 were retained to remove rare tokens and tokens that dominate the dataset, resulting in 9,265 unique words in the vocabulary. (1).

## Clustering Methodology

A co-occurrence matrix was constructed using a sliding window of 5 words and normalized with L2 normalization to highlight relative co-occurrences. Dimensionality reduction was applied in two stages: PCA reduced the matrix to 20 dimensions, preserving most variance, followed by UMAP for further reduction to 5 dimensions. The hyperparameters of UMAP, the number of neighbours, the minimum distance and the output dimensions, were optimized by using grid search to balance the preservation of local and global structures (2). K-means clustering was then applied for  $K$  values ranging from 2 to 20, with configurations evaluated using Silhouette Scores (measuring cohesion and separation) and Inertia (assessing compactness) (3).

## Applying Occam's Razor

The Occam's Razor principle was used to identify that  $K = 7$  was the optimal number of clusters based on previous preprocessing. At higher  $K$  values, the inertia dropped as the clusters became more compact, however, it came at the cost of lower interpretability, overfitting as well as low generalizability. In addition to this, the increase in silhouette score seemed to plateau. As a result, a compromise was made with model performance and complexity, and  $K$  was selected to be 7 as this value effectively captured the patterns in the data with high silhouette scores (relatively) and inertia score without unnecessary complexity (4).

## Results and Analysis

After evaluating performance,  $K = 7$  was the optimal number of clusters, capturing meaningful word co-occurrences while balancing complexity and generalization. The validation

Silhouette Score was 0.698, and the test score was 0.633, with minimal degradation on unseen data. The inertia appeared large due to the application of UMAP on the dataset. However, the elbow method was still used to determine the elbow of the graph at  $K = 7$ .

| $K$ | Silhouette Score | Inertia   |
|-----|------------------|-----------|
| 3   | 0.611            | 40213.301 |
| 4   | 0.632            | 33751.602 |
| 5   | 0.653            | 29646.887 |
| 6   | 0.661            | 25722.094 |
| 7   | 0.698            | 23330.656 |

Table 1: Clustering Results for Different  $K$  Values

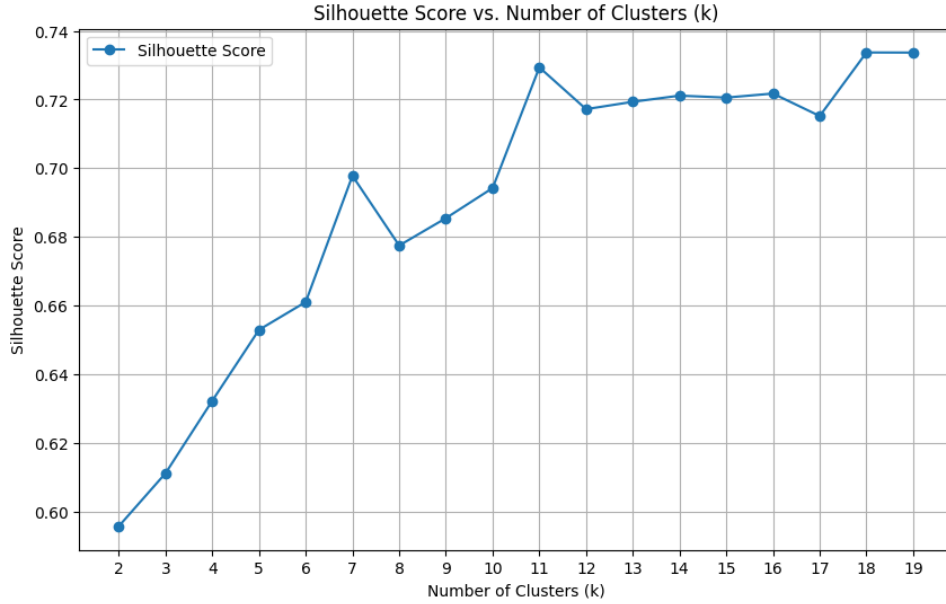


Figure 1: Silhouette Score vs.  $K$ . Optimal performance achieved at  $K = 4$ .

## References

- [1] Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
- [2] McInnes, L., Healy, J., & Melville, J. (2018). UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction. *arXiv preprint arXiv:1802.03426*.
- [3] Arthur, D., & Vassilvitskii, S. (2007). K-means++: The Advantages of Careful Seeding. *Proceedings of the 18th Annual ACM-SIAM Symposium on Discrete Algorithms*.
- [4] Domingos, P. (1999). The Role of Occam’s Razor in Knowledge Discovery. *Data Mining and Knowledge Discovery*, 3(4), 409–425.